

MEZRAIT LIUL MENBER

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Research interests: verifying security claims under deployment conditions; cryptographic analysis and empirical measurement; systems security

EDUCATION

Abu Dhabi University

Abu Dhabi, UAE

B.Sc. in Cybersecurity Engineering

Expected June 2027

- Dean's List: **five consecutive** semesters (Fall 2023 – Fall 2025); full undergraduate scholarship awarded by the Government of the United Arab Emirates
- Relevant Coursework: Introduction to Cryptography; Network Security and Forensics; Ethical Hacking; Information Security; Digital Forensics; Data Structures and Algorithms; Discrete Mathematics; Statistics and Research Methods
- Selected course projects: [NIST data-leakage digital-forensics case study](#); [Abu Dhabi Islamic Bank cybersecurity risk & resilience audit](#)

PUBLICATIONS

AEX-128: A Key-Dependent Finite-State SPN–ARX Hybrid Block Cipher — *IEEE Access*, 2026

- M. Alawida, **M. L. Menber**, W. H. Alshoura, R. Almajed, S. Almuhammadi — doi: [10.1109/ACCESS.2026.3730739](https://doi.org/10.1109/ACCESS.2026.3730739)
- A 128-bit cipher whose round structure is selected at runtime by a key-dependent finite-state machine, yielding **59,049** distinct execution paths — evaluated via MILP-based active-S-box analysis, SAC chi-square testing, and NIST SP 800-22 at 10^6 -bit scale

OTHER RESEARCH

Odd Weird Numbers: Prime-Factor Bounds for Near-Perfect Equations

Submitted, J. Number Theory, 2026

- Independent single-author research in computational/analytic number theory, outside my cryptography collaboration: proved two prime-factor lower bounds for near-perfect numbers and a sharpened numerical bound (to 6.09×10^{12}) via an exact two-stage branch-and-bound search with certified proof traces, implemented in Python with exact rational arithmetic

RESEARCH EXPERIENCE

Undergraduate Researcher, Block Cipher Design

Abu Dhabi, UAE

Abu Dhabi University — Supervisor: Dr. Moatsum Alawida

Feb 2026 – Present

- Co-designed AEX-128, a published 128-bit block cipher combining SPN and ARX rounds under a key-dependent finite-state machine — **3^{10} (59,049)** possible execution paths
- Designed the ChaCha-based key schedule: 512-bit state expansion via quarter-round mixing deriving 12 subkeys and per-round FSM transition parameters from key entropy
- Built the Java reference implementation used for the paper's benchmarks — **0.58 μ s/block** encryption — and designed the initial diffusion experiments, including the avalanche methodology used in the published paper
- Co-developed the security evaluation with my supervisor — MILP-based active-S-box analysis across all 59,049 FSM execution paths, SAC testing with chi-square analysis (**94.5%** pass at 95% confidence), and NIST SP 800-22 testing at 10^6 -bit scale — and have since rebuilt the MILP analysis independently (see Projects)

TECHNICAL SKILLS

Cryptography & analysis: block cipher design, MILP active-S-box analysis (PuLP/HiGHS; independently reproduced), SAC/chi-square and NIST SP 800-22 statistical testing, ChaCha-based key scheduling

Systems & backend: Python, FastAPI, PostgreSQL with Alembic migrations, Docker, MCP (Model Context Protocol), Linux, headless-Chromium measurement

Frontend: TypeScript, React/TSX

Also familiar: Java (cipher reference implementation), C, SQL; Git/GitHub, automated testing, LightRAG and knowledge-graph retrieval

PROFESSIONAL EXPERIENCE

Python Developer Intern

Abu Dhabi, UAE

Ansen Technology Co. Ltd. — AI-security company, eThreats threat-intelligence platform

29 Jun – 24 Aug 2026

- Replaced the document-retrieval pipeline with a knowledge-graph backend (**LightRAG + Apache AGE**) fronted by an adapter layer, rolled out via shadow ingest in parallel with the legacy system; answer faithfulness on a 15-query golden set rose from **86.7% to 100%** and p99 latency improved, with ingest-throughput and p50/p95 regressions reported alongside the gains
- Traced **94.1%** of raw tool-call failures in a live skill-on/off experiment to a single namespace mismatch in the sources service's capability signal, and fixed it; separately widened the MCP gateway cache from per-task to per-run scope, eliminating redundant lookups in **294 of 385** historical runs — both shipped with regression tests
- Built the agent skill-rating lifecycle end-to-end (evidence → draft → human review → active) across backend and frontend, including the Skill Reviewer surface in the analyst workspace
- Contributed to the slide-generation pipeline (seven archetype templates, a measurement core that moved quality validation from the authored design file to the exported deck itself, bounded failure paths after two leaked-browser-process incidents) and ran a branch-wide test-coverage sweep that surfaced four production bugs earlier review had missed

Course Assistant & Peer Mentor

Abu Dhabi, UAE

Abu Dhabi University

Jun 2024 – Present

- **250+** hours tutoring across math, physics, chemistry, and engineering (five consecutive semesters); mentored **50+** first-year students; peer-mentorship certificate each term

PROJECTS

AEX-128 MILP Analysis — Independent Reproduction | *Python, PuLP/HiGHS* github.com/Mezrait/aex128-milp-reproduction

- Rebuilt the byte-truncated differential MILP model from scratch; exhaustively solved all **59,049** execution paths to proven optimality, with a solver-free dynamic-programming cross-verification
- Validated the setup by independently reproducing the classical 5-round AES bound of **26** active S-boxes

Secure Collaborative Cyber Threat Intelligence Platform | Capstone, team of 4; Supervisor: [Dr. Abid Mehmood](#) *Fall 2026 – in build*

- Three-tier platform for cross-organization sharing of indicators of compromise under NIST SP 800-207 zero-trust constraints: multi-stage human-verified intel pipeline, OWASP-based threat model, cryptographically hash-chained append-only audit log for tamper evidence, MITRE ATT&CK mapping, organization-scoped visibility (private / org / partner / community)

HONORS & AWARDS

GISEC 2025: 5th place (of 10+ finalist teams), School of Cyber Defense CTF — Dubai Electronic Security Center; qualified 2nd among university teams in round one

Sharjah Voluntary Award: 2024 & 2025 consecutive recipient; 450+ hours of university community service

Languages: Amharic (native); English (fluent)